CS2610: Computer Organization and Architecture Lab

Lab 1: Introduction to RISC-V Toolchain and Assembly

2nd February 2026

Objective

In this lab, you will be introduced to writing basic RISC-V assembly programs. The primary goal of this lab is to familiarize you with fundamental RISC-V assembly instructions and the RISC-V toolchain.

1 Question 1

Write a RISC-V assembly program that loads two word-sized integers A and B from memory and computes the following values:

1. $A + B$
2. $A - B$
3. A `AND` B
4. A `OR` B
5. (`NOT` A) `XOR` (`B`)

Store all five results in **consecutive memory locations**.
Using `spike` commands, perform the following:

- Dump the contents of the two input memory locations at the start of execution
- Dump the contents of all result memory locations at the end of execution

2 Question 2

Write a RISC-V assembly program that initializes an array of 5 integers in memory. Using a loop, modify the array as follows:

- If the index of the element is even, add 1 to the element
- If the index of the element is odd, subtract 1 from the element

Assume **0-based indexing**.

Example:

```
Initial array: [10, 10, 10, 10, 10]
Final array:   [11,  9, 11,  9, 11]
```

Using `spike` commands, perform the following:

- Dump the contents of the array before execution
- Dump the contents of the array after execution
- Dump the contents of the array after first 3 loop iterations

3 Submission Instructions

1. Have a chat with your TA
2. The assignment must be completed individually.
3. The following artifacts must be submitted:
   - Screenshots of the outputs as specified for each question
   - Code files for each question
4. Compress all files into a single ZIP archive and submit it on Moodle.